

The Himalayan Tightrope

Balancing the Economy-Environment Trade-off

Ritwiz Sarma and Riju Garg

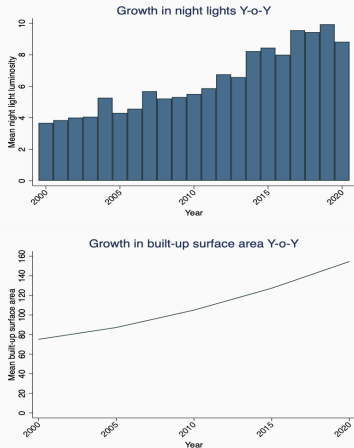
March 4, 2025

Madras School of Economics

Presented at the 59th Annual Conference of the Indian Econometric Society.

The Trade-off Exists...

- Why build infrastructure?
 - Network effects, economic growth
 - Administrative, defense reasons
 - Political incentives



The Trade-off Exists...

- Why build infrastructure?
 - Network effects, economic growth
 - Administrative, defense reasons
 - Political incentives
- But then, there is a growing *long-run* and *human* cost.

Landslides biggest killer in Uttarakhand; highest no. of casualties this year

Himachal cloudbursts: Are hydropower projects to blame?

Four of the 6 places where cloudbursts occurred last week were locations of hydro projects

Man-made disaster in Himachal Pradesh

The result: Monsoon showers triggered flash floods in riverside towns and highways in July and August and caused massive loss of life and property.

Himachal: 4 injured as landslide triggered by rain hits bus in Manali, markets flooded in Kullu

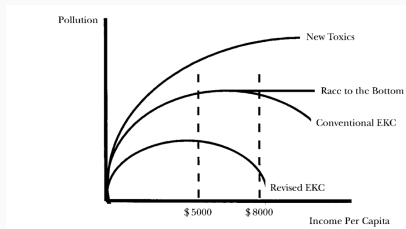
By Shailee Dogra ✕

Frequent Natural Disasters Continue To Haunt Himachal Pradesh, Scientists Call For Urgent 'Risk Avoidance'

... But Do We Measure It Right?

Environmental Kuznets Curve

- Inverted-U relationship between economic growth and environmental degradation.
- Contradictory findings in recent literature.



Different EKC adaptations from Dasgupta et al. (2002).

- How does *infrastructure-led* growth relate to environmental degradation?
- Does the Environmental Kuznets curve hold?
- Do macro-level models hold for:
 - sensitive ecologies?
 - micro-level data?

Our answer: Not well.

- We build a high-res geospatial dataset at village/town level
- We estimate the EKC: 5 indicators, 20 years of data
- We focus on two Himalayan states, where incentives exist for both infrastructure investment and ecological sustainability

Data

Shrid Usually single village/town, can be aggregation if merged/separated between 1990-2013.

Indices Standardization, then take the:

- arithmetic mean, or
- principal component

Night lights

Why? Strong correlation with GDP growth even at subnational level (Doll, Muller, and Morley, 2006).

Source NASA, harmonized across satellites by Li, Zhou, and Zhao (2020), yearly data.

Night lights

Why? Strong correlation with GDP growth even at subnational level (Doll, Muller, and Morley, 2006).

Source NASA, harmonized across satellites by Li, Zhou, and Zhao (2020), yearly data.

Built-up area

Why? Detects economic activity at highly local levels (Baragwanath et al., 2021).

Source ESA, Copernicus GHSL product, five-year intervals.

PM2.5 air quality

Why? Respected measure of environmental degradation (Caviglia-Harris, Chambers, and Kahn, 2009).

Source SHRUG v2, Development Data Lab cf. Donkelaar et al. (2021).

PM2.5 air quality

Why? Respected measure of environmental degradation (Caviglia-Harris, Chambers, and Kahn, 2009).

Source SHRUG v2, Development Data Lab cf. Donkelaar et al. (2021).

Land surface temperature

Why? Long-run changes from environmental degradation, global warming (European Space Agency, n.d.).

Source NASA MODIS, yearly aggregates from 8-weekly data.

Forest cover

Why? Relevant for infra growth, also used for horizontal devolution (Wang, 2003; XV Finance Commission, 2020).

Source SHRUG v2, Development Data Lab, Terra MODIS (NASA).

Model and Findings

General Quadratic Model

$$\text{Econ}_{it} = \beta_0 + \beta_1(\text{Env}_{it}) + \beta_1(\text{Env}_{it}^2) + \eta_i + \gamma_t + \epsilon_{it}$$

where:

- Econ is the economic index variable.
- Env is the environmental index variable.
- η_i are *shrid*-level fixed effects.
- γ_i are year fixed effects.
- ϵ_{it} is the idiosyncratic error term.

- Investigates the trade-off between economic growth and environmental sustainability.
- Uses high-resolution geospatial data for Himachal Pradesh and Uttarakhand.
- Develops a novel dataset of economic and environmental indicators over two decades.

Context: Why Build Infrastructure?

- Economic, political, and administrative motivations.
- Border security concerns with China, Nepal, and Bhutan.
- Economic transition from agriculture to manufacturing and services.
- Role of infrastructure in connectivity, digital access, and energy generation.

Theoretical Framework: Environmental Kuznets Curve (EKC)

- Hypothesis: Inverted-U relationship between economic growth and environmental degradation.
- Early stages: Growth leads to increased degradation.
- Later stages: Growth enables sustainability efforts.
- Contradictory findings in recent literature.

- Study unit: SHRUG village/town-level data.
- Economic indicators: Night-time lights, built-up area.
- Environmental indicators: Air quality (PM2.5), forest cover, land surface temperature.
- Data sources: NASA MODIS, GHSL, VIIRS, DMSP-OLS.

Findings: Temporal Shifts in Himachal Pradesh

- Early 2000s - Mid 2000s: Forest cover increases in northern districts, declines in southern regions.
- 2010s: Growth in night-time lights, economic activity expands.

Findings: Temporal Shifts in Uttarakhand

- Early 2000s - Mid 2000s: Forest cover increases in the north, declines in the south.
- Mid 2000s - 2010s: Widespread forest cover loss.
- Post-2010: Recovery in forest cover.
- Night-time lights show growth across all periods.

- Quadratic fixed-effects model:
- Economic Index (NTL, built-up area) vs. Environmental Index (PM2.5, forest cover, LST).
- Fixed effects used based on Hausman test.

Regression Results: First Epoch

- Estimated relationship follows the EKC (inverted U-shape).
- Economic growth initially leads to environmental degradation, then stabilizes.

Regression Results: Second Epoch

- Relationship reverses to a U-shape.
- Economic growth initially reduces degradation but later exacerbates it.

- Results suggest heterogeneity across time.
- EKC not stable in ecologically sensitive regions.
- Potential tipping point in Himalayan environmental dynamics.

- Need for localized, dynamic policy frameworks.
- Infrastructure planning should integrate ecological risk assessments.
- Emphasis on renewable energy, sustainable urban planning.

- Economy-environment trade-offs in the Himalayas are complex.
- EKC model does not universally apply.
- Future research: pinpoint specific projects to build a treatment group for causal analysis.

Thank you.

Open to questions/comments.